

Exp : 6

16 bit subtraction

AIM:

To write an assembly language program to implement 16 bit subtraction using 8085 processor.

ALGORITHM:

- 1) Start the program by loading a register pair with address of 1st number.
- 2) Copy the data to another register pair.
- 3) Load the second number to the first register pair.
- 4) Sub the two register pair contents.
- 5) Check for carry.
- 6) Store the value of sub and carry in memory locations.
- 7) End.

PROGRAM:

MNEMONICS	Explanations :
LHLD 2050	Load the 16 bit data from memory location 2050H and 2051H into the HL register pair.
XCHG.	Exchange the contents of the HL and DE register pairs.
LHLD 2052	Load the 16-bit data from 2052H and 2053H into HL register pair.
MVI C, 00	Move immediate value 00H to C.

MOV A, E

move the content of register E into A.

SUB L

subtract the content of L from A.

STA 2054

store result of subtraction into 2054 H.

MOV A, D

move the content of D to A.

SUB H

subtract content of H from A.

STA 2055

store the result of sub. into 2055 H.

HLT

Halt the execution of program.

INPUT

Address

Data

2050

2

2052

3

OUTPUT

Address

Data

017 2054

1

2055

1

RESULT:

Thus the program was executed successfully using 8085 processor simulator.



Registers

A	00
BC	05 00
DE	00 03
HL	00 02
PSW	00 00
PC	42 14
SP	FF FF
Int-Reg	00

Flag

S	0
Z	1
AC	0
P	1
C	0

Load me at

```
1  LHLD 2050
2  XCHG
3  LHLD 2052
4  MVI C,00
5  MOV A,E
6  SUB L
7  STA 2054
8  MOV A,D
9  SUB H
10 STA 2055
11 HLT
```

Decimal - Hex Conversion

Decimal Hex

0	0
→ To Hex	← To Dec

I/O Ports

0	-	+	00
Update Port Value			

Memory

2052	-	+	02
Update Memory			

Data Stack KeyPad Memory I/O Ports

Start 2054

OK

Address (Hex)	Address	Data
0806	2054	1
0807	2055	0
0808	2056	0
0809	2057	0
080A	2058	0
080B	2059	0
080C	2060	0
080D	2061	0
080E	2062	0
080F	2063	0
0810	2064	0
0811	2065	0
0812	2066	0
0813	2067	0
0814	2068	0
0815	2069	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

